Model-based (parametric) spectral estimation

Signal models

zero-mean white noise $e(n)$ $E[e^2(n)] = \sigma_e^2 = 1$ → $\mathbf{H(z)}$ → N samples of observed signal $x(n)$

AR(p) (autoregressive) :

$$x(n) + \sum_{i=1}^{p} a_i\, x(n-i) = b_0\, e(n)$$
$$H(z) = \frac{b_0}{1+\sum_{i=1}^{p} a_i\, z^{-i}}$$

MA(q) (moving average)

$$x(n) = \sum_{i=0}^{q} b_i\, e(n-i)$$
$$H(z) = \sum_{i=0}^{q} b_i\, z^{-i}$$

ARMA(p,q) (autoregressive moving average)

$$x(n) + \sum_{i=1}^{p} a_i\, x(n-i) = \sum_{i=0}^{q} b_i\, e(n-i)$$
$$H(z) = \frac{\sum_{i=0}^{q} b_i\, z^{-i}}{1+\sum_{i=1}^{p} a_i\, z^{-i}}$$

General method

- Given N samples of observed signal $x(n)$
- Estimate $\hat{r}_x(k) = \frac{1}{N}\sum_{n=0}^{N-1} x(n)\, x(n+k)$
- Assume orders (p, q) (# of poles / zeroes)
- Using $\hat{r}_x(k)$, find $\{\hat{a}_i\}_{i=1,\dots,p}$, $\{\hat{b}_i\}_{i=1,\dots,q}$ from relationship $\{a_i\}_{i=1,\dots,p}$, $\{b_i\}_{i=1,\dots,q} \rightarrow r_x(k)$
- $\{\hat{a}_i\}_{i=1,\dots,p}$ and $\{\hat{b}_i\}_{i=1,\dots,q}$ gives $\hat{H}(z)$ which gives $\hat{H}(f)$
- Then $\hat{P}_x(f) = \sigma_e^2\, |\hat{H}(f)|^2 = |\hat{H}(f)|^2$

AR(p) details

Yule-Walker equations: $\begin{bmatrix} \hat{r}_x(0) & \cdots & \hat{r}_x(p-1) \\ \vdots & \ddots & \vdots \\ \hat{r}_x(p-1) & \cdots & \hat{r}_x(0) \end{bmatrix} \begin{bmatrix} \hat{a}_1 \\ \dots \\ \hat{a}_p \end{bmatrix} = -\begin{bmatrix} \hat{r}_x(1) \\ \vdots \\ \hat{r}_x(p) \end{bmatrix}$, $\quad \hat{b}_0^2 = \hat{r}_x(0) + \sum_{k=1}^{p} \hat{a}_k\, \hat{r}_x(k)$

Solution of Yule-Walker equations:

- YW-method: solve equations directly, **MATLAB**: ***aryule*** → $\{a_i\}$, ***pyulear*** → $\hat{P}_x(f)$
- Covariance method, **MATLAB**: ***arcov*** → $\{a_I\}$, ***pcov*** → $\hat{P}_x(f)$
- Modified covariance method, **MATLAB**: ***armcov*** → $\{a_I\}$, ***pmcov*** → $\hat{P}_x(f)$
- Burg method, **MATLAB**: ***arburg*** → $\{a_i\}$, ***pburg***→ $\hat{P}_x(f)$

Model order estimation: minimize $\mathrm{AIC}(p) = N \log \hat{\varepsilon}(p) + 2p$ or $\mathrm{MDL}(p) = N \log \hat{\varepsilon}(p) + p \log N$
where the estimated model error is $\hat{\varepsilon}(p) = \hat{b}_0^2 = \hat{r}_x(0) + \sum_{k=1}^{p} \hat{a}_k\, \hat{r}_x(k)$

AR Methods

	Burg	Covariance	Modified Covariance	Yule-Walker
Characteristics	Does not apply window to data	Does not apply window to data	Does not apply window to data	Applies window to data
	Minimizes the forward and backward prediction errors in the least squares sense, with the AR coefficients constrained to satisfy the L-D recursion	Minimizes the forward prediction error in the least squares sense	Minimizes the forward and backward prediction errors in the least squares sense	Minimizes the forward prediction error in the least squares sense (also called "Autocorrelation method")
Advantages	High resolution for short data records	Better resolution than Y-W for short data records (more accurate estimates)	High resolution for short data records	Performs as well as other methods for large data records
	Always produces a stable model	Able to extract frequencies from data consisting of *p* or more pure sinusoids	Able to extract frequencies from data consisting of *p* or more pure sinusoids	Always produces a stable model
			Does not suffer spectral line-splitting	
Disadvantages	Peak locations highly dependent on initial phase	May produce unstable models	May produce unstable models	Performs relatively poorly for short data records
	May suffer spectral line-splitting for sinusoids in noise, or when order is very large	Frequency bias for estimates of sinusoids in noise	Peak locations slightly dependent on initial phase	Frequency bias for estimates of sinusoids in noise
	Frequency bias for estimates of sinusoids in noise		Minor frequency bias for estimates of sinusoids in noise	
Conditions for Nonsingularity		Order must be less than or equal to half the input frame size	Order must be less than or equal to 2/3 the input frame size	Because of the biased estimate, the autocorrelation matrix is guaranteed to positive-definite, hence nonsingular

MA(q) details

- Yule-Walker equations becomes non-linear in $\{\hat{b}_i\}_{i=1,\ldots,q}$
- Methods exist to estimate $\{\hat{b}_i\}_{i=1,\ldots,q}$ from $x(n)$ directly or from $\hat{r}_x(k)$
  MATLAB: ***armax***

- However: since the MA(q) model corresponds to a q+1 tap FIR-filter
  $r_x(k) = 0, \ |k| > q$
- Therefore an equivalent and simpler approach is to use the (non-parametric) Blackman-Tukey estimate with a rectangular window of width 2q+1:
  $\hat{P}_x(f) = \sum_{k=-q}^{q} \hat{r}_x(k)\, e^{-j2\pi \frac{f}{f_s} k}$

ARMA(p,q) details

- Yule-Walker equations becomes non-linear in $\{\hat{a}_i\}_{i=1,\ldots,p}$, $\{\hat{b}_i\}_{i=1,\ldots,q}$
- Methods exist to estimate $\{\hat{a}_i\}_{i=1,\ldots,p}$, $\{\hat{b}_i\}_{i=1,\ldots,q}$ from $x(n)$ directly or from $\hat{r}_x(k)$
  MATLAB: ***armax***

- However a simpler three-step approach can be used:

- First, find $\{\hat{a}_i\}_{i=1,\ldots,p}$ from MYWE, Modified Yule-Walker equations
  $$\begin{bmatrix} \hat{r}_x(q) & \cdots & \hat{r}_x(q-(p-1)) \\ \vdots & \ddots & \vdots \\ \hat{r}_x(q+(p-1)) & \cdots & \hat{r}_x(q) \end{bmatrix} \begin{bmatrix} \hat{a}_1 \\ \ldots \\ \hat{a}_p \end{bmatrix} = - \begin{bmatrix} \hat{r}_x(q+1) \\ \vdots \\ \hat{r}_x(q+p) \end{bmatrix}$$

  This gives $\hat{P}_{x,denom}(f) = \left| 1 + \sum_{i=1}^{p} \hat{a}_i\, e^{-j2\pi \frac{f}{f_s} i} \right|^2$

- Secondly, filter $x(n)$ with $G(z) = \text{denom}\{H(z)\} = 1 + \sum_{i=1}^{p} a_i\, z^{-i}$
  giving a MA(q) signal, $y(n)$

- Thus, finally, estimate $\hat{r}_y(k)$ and find $\hat{P}_{x,nom}(f) = \sum_{k=-q}^{q} \hat{r}_y(k)\, e^{-j2\pi \frac{f}{f_s} k}$ as for MA(q)
  Then $\hat{P}_x(f) = \frac{\hat{P}_{x,nom}(f)}{\hat{P}_{x,denom}(f)}$
